Architecting the Autonomous Video Factory: A Comprehensive Technical Analysis of State-of-the-Art Open Source Models for 16GB VRAM Environments (2025-2026)

1. Introduction: The Democratization of Automated Content Production

The landscape of artificial intelligence has undergone a seismic shift as we transition through 2025 into 2026. We are witnessing the collapse of the barrier between enterprise-grade capability and consumer-grade hardware. For years, the domain of high-fidelity video analysis and generation was the exclusive purview of massive GPU clusters, requiring compute resources that were financially and logistically out of reach for individual creators or small studios. However, a confluence of architectural innovations—specifically in quantization, efficient attention mechanisms, and specialized "small" language models—has brought these capabilities down to the desktop workstation. The focal point of this revolution is the enablement of the "Autonomous Video Shorts Factory," a fully automated pipeline capable of ingesting raw long-form footage, understanding its semantic and temporal context, identifying viral moments, tracking subjects for vertical reformatting, transcribing speech with phoneme-level precision, and generating supplemental visual assets to produce a polished, platform-ready video short.

This report serves as a definitive technical blueprint for constructing such a factory within the strict constraints of a single NVIDIA RTX 5080 GPU, equipped with 16GB of GDDR7 VRAM. The selection of the RTX 5080 as the target hardware is deliberate; it represents the upper echelon of the consumer market—accessible yet constrained enough to require rigorous architectural discipline. Unlike the data center A100 or H100 GPUs with their expansive 80GB memory banks, the 16GB limit of the RTX 5080 forces a shift from monolithic model deployment to intelligent, sequential resource orchestration. It necessitates a "Load-Infer-Unload" architecture where models are treated as transient agents rather than persistent services.

The analysis presented herein is exhaustive, drawing upon the latest developments in open-source AI as of early 2026. We dissect the specific VRAM requirements of leading models, evaluate their performance on state-of-the-art benchmarks like Video-MME and Open ASR, and synthesize these findings into a cohesive system architecture. We explore how

models like **Qwen2.5-VL**, **SAM 2**, **NVIDIA Canary**, and **Wan 2.1** can be harmonized to operate within this memory envelope without compromising the quality that audiences demand. Furthermore, we delve into the emerging paradigm of "Agentic Editing," where Vision-Language Models (VLMs) do not merely classify content but actively direct the editing process, outputting structured execution plans that drive downstream tracking and generation tools. This move from passive analysis to active orchestration marks the defining characteristic of the 2025-2026 autonomous workflow.

2. Hardware Constraints and Architectural Strategy

The primary challenge in designing an autonomous video factory on consumer hardware is the finite VRAM budget. The RTX 5080, while offering substantial compute throughput via its Ada Lovelace or Blackwell-derived architecture, is bound by its 16GB frame buffer. In the context of modern AI, 16GB is a "tight" environment when dealing with multimodal tasks. A single unquantized 7B parameter model in 16-bit precision (FP16) requires approximately 14-15GB of VRAM just for weights and minimal inference overhead, leaving no room for the KV cache required for long context windows or for loading secondary models.

Therefore, the factory cannot be architected as a set of simultaneous microservices. Instead, it must function as a serialized assembly line. The orchestration logic—likely a Python-based state machine—must rigorously manage the lifecycle of each model. The pipeline operates in distinct phases: Analysis, Audio Processing, Spatial Tracking, and Generation. Each phase has a strict "VRAM Budget" that dictates which models can be loaded. This constraint elevates **Quantization** from an optimization technique to a structural necessity. By 2025, techniques like **GGUF (GPT-Generated Unified Format)** and **AWQ (Activation-aware Weight Quantization)** have matured to the point where 4-bit quantization offers a perplexity degradation so minimal that it is imperceptible for most "instruction-following" tasks required in video editing.

This report operates under the premise that all Large Language Models (LLMs) and VLMs will be deployed in 4-bit or 8-bit quantized formats, while vision encoders and diffusion UNets—which are far more sensitive to quantization noise—will largely remain in FP16 or BF16 to preserve visual fidelity. The optimization goal is to fit the active working set into roughly 12-14GB of VRAM, leaving 2-4GB for the Operating System (OS), display drivers, and transient spikes in activation memory during inference.

3. The Director Agent: Video Understanding and Viral Moment Detection

The first and most critical station in the factory is the "Director Agent." This component is responsible for watching the raw footage—which could be an hour-long podcast, a gaming stream, or a vlog—and making high-level editorial decisions. It must identify segments that are not just coherent, but *engaging*. This requires a model with deep temporal reasoning, the

ability to understand humor and sarcasm, and precise timestamping capabilities.

3.1 The Rise of Temporal-Aware Vision Language Models

In the period of 2023-2024, video understanding was often hacked together by treating video as a sequence of disjointed images fed into an Image-LLM. This approach failed to capture the *flow* of events. The 2025-2026 generation of models, however, introduces native temporal understanding through architectural innovations like **Multimodal Rotary Positional Embeddings (M-RoPE)**. This technology allows the model to encode the absolute temporal position of visual tokens, giving it an innate sense of "when" an event occurs relative to the video's timeline.

3.2 Qwen2.5-VL: The Benchmark Leader

The analysis identifies **Qwen2.5-VL-7B-Instruct** as the paramount choice for the 16GB constraint.[1] Released in early 2025 by the Qwen team, this model represents a significant leap over its predecessors. Its defining feature is **Naive Dynamic Resolution**, which dynamically maps video frames to visual tokens based on information density. Unlike models that force every frame into a fixed 336x336 patch grid, Qwen2.5-VL allocates more tokens to complex frames (e.g., a screen showing text) and fewer to simple ones (e.g., a static background), optimizing the context window usage.[2]

Crucially, Qwen2.5-VL supports context lengths exceeding 1 hour of video, a capability previously reserved for proprietary models like Gemini 1.5 Pro.[2] Benchmarks reinforce its dominance; on the **Video-MME** leaderboard, a rigorous test of long-context multimodal understanding, Qwen2.5-VL-7B achieves scores that rival much larger models, demonstrating robust temporal reasoning.[2] It excels in "Agentic" tasks, meaning it can follow complex instructions to output structured JSON data—a mandatory requirement for an automated pipeline. For example, it can be prompted to "Identify the three most funny moments, return their start and end timestamps, and provide a description of the main subject for tracking."

VRAM Analysis for Qwen2.5-VL:
Running the 7B model in its native BF16 format would consume roughly 15-16GB, dangerously close to the RTX 5080's limit. However, the community has rapidly adopted AWQ and GGUF quantization for this architecture. The Qwen2.5-VL-7B-Instruct-AWQ (Int4) variant reduces the model footprint to approximately 5.5 - 6.5 GB.4 This is the "sweet spot." It leaves roughly 10GB of headroom for the KV cache (which grows with video length) and system overhead. Users on 8GB cards struggle with this model, often facing OOM errors with long contexts, but for the 16GB RTX 5080, it fits comfortably, allowing for the ingestion of substantial video buffers without paging to system RAM.

3.3 Competitive Alternatives: InternVL 2.5 and MiniCPM-V

While Qwen leads, the ecosystem is rich with alternatives. **InternVL 2.5-8B**, released in late 2024/early 2025, utilizes a "Mixture of Experts" (MoE) inspired approach to scaling.[6] It boasts

a massive 6B parameter vision encoder (InternViT-6B), which gives it exceptional visual acuity—perhaps superior to Qwen for recognizing fine details in static images. However, this large vision backbone is heavy and less amenable to quantization without accuracy loss.[8] For a video pipeline where *temporal* reasoning (the LLM's job) is often more important than pixel-peeping, the heavy vision encoder of InternVL makes it a second-tier choice for the memory-constrained 5080 compared to the balanced architecture of Qwen.

MiniCPM-V 2.6 (8B) is another strong contender, specifically optimized for edge devices.[9] It features "state-of-the-art token density," encoding images into significantly fewer tokens than competitors. This makes it the fastest option for processing high-framerate video. However, its performance on complex reasoning benchmarks like MMMU (50.4%) trails behind Qwen2.5-VL-7B (58.6%) and InternVL 2.5 (70%+).[2] For a "Shorts Factory" where understanding the *joke* or the *drama* is key, the deeper reasoning capabilities of Qwen take precedence over the raw speed of MiniCPM.

3.4 Operationalizing the Director Agent

In the factory workflow, Qwen2.5-VL is loaded first. It consumes the video features—often pre-extracted or streamed—and generates a JSON manifest. This manifest contains the "Edit Decision List" (EDL): timestamps for cuts, bounding box coordinates for the initial tracking target, and text prompts for downstream generation. Once this JSON is generated and validated, Qwen is unloaded from VRAM to make way for the next set of specialist agents.

4. The Cameraman Agent: Intelligent Visual Tracking and Reframing

Once the viral segment is identified, the challenge shifts to formatting. Converting 16:9 landscape footage to 9:16 vertical video for TikTok or Reels is not a simple crop operation. It requires an active "Cameraman" agent that can identify the subject of interest (the speaker, the ball, the action) and pan the virtual camera to keep them centered. This domain is dominated by object tracking and segmentation models.

4.1 SAM 2: The Unified Segmentation and Tracking Model

Segment Anything Model 2 (SAM 2), developed by Meta, stands as the gold standard for this task in 2025-2026.[12] Unlike its predecessor, SAM 2 is natively designed for video. It introduces a **streaming memory architecture** consisting of a memory bank and a memory attention module.[13] This allows the model to "remember" the object it is tracking across frames, even if the object is temporarily occluded or turns away from the camera.

The primary advantage of SAM 2 is its **Zero-Shot Generalization**. The Director Agent (Qwen) provides a text description or a bounding box for the first frame (e.g., "the person in the red shirt"). SAM 2 takes this prompt and propagates the segmentation mask through the entire

video clip.[14] This propagation is robust to diverse visual domains, from gaming interfaces to real-world vlogs.

VRAM Optimization for SAM 2:
Historically, VOS (Video Object Segmentation) models were VRAM hogs, as they cached feature maps for every frame. However, optimization techniques codified in projects like Det-SAM2 and official updates have introduced mechanisms to cap memory usage.15 By setting a fixed size for the "memory bank" (e.g., remembering only the last N frames plus the initial frame) and offloading the bulk of the video data to system RAM (CPU), SAM 2 can run inference on infinitely long videos with a constant VRAM footprint. For the "Small" or "Tiny" variants of SAM 2—which are sufficient for determining crop coordinates—the VRAM usage is remarkably low, often staying under 2GB.16 This efficiency allows it to potentially coexist in VRAM with the ASR model during the middle phase of the pipeline.

4.2 High-Speed Alternatives: YOLO-World and CAMELTrack

While SAM 2 offers pixel-perfect masks (useful for background replacement effects), sometimes raw speed is preferred. **YOLO-World v2** represents the state-of-the-art in open-vocabulary object detection.[17] Unlike traditional YOLO models trained on fixed classes (like COCO's 80 objects), YOLO-World can detect any object described by text. It is incredibly fast, capable of running at over 60 FPS on an RTX 5080.

To turn detection into tracking, YOLO-World is often paired with a tracker like **CAMELTrack** (Context-Aware Multi-cue ExpLoitation).[19] CAMELTrack, a 2025 benchmark winner, uses transformer modules to associate detections across frames, handling complex motion better than heuristic trackers like ByteTrack.[20] For a basic "keep the face centered" task, the YOLO-World + CAMELTrack combination is lighter and faster than SAM 2, consuming less than **1GB VRAM**. However, SAM 2 is recommended for the "Factory" because its segmentation masks enable secondary creative effects (like placing text behind the subject) that add production value to the final short.

5. The Scribe Agent: ASR and Temporal Alignment

Viral shorts are unwatchable without captions. The "Scribe Agent" must transcribe speech with near-perfect accuracy and, crucially, align that text to the audio with millisecond precision to enable dynamic, word-by-word caption animations ("karaoke style").

5.1 NVIDIA Canary: The New ASR Benchmark

For years, OpenAI's Whisper was the default choice. However, by 2025, **NVIDIA's Canary-1B** and its successor **Canary-1B-v2** have emerged as the superior option for open-source deployment.[21] Canary tops the Open ASR Leaderboard, delivering a Word Error Rate (WER) of approximately 6.67% across a wide range of languages, significantly outperforming Whisper Large V3 while being 3x smaller.[23]

Canary-1B utilizes a **FastConformer** encoder and a Transformer decoder, an architecture optimized for efficiency on NVIDIA silicon.[24] It supports automatic punctuation and capitalization (PnC) natively, which is essential for readability. Furthermore, it supports bi-directional translation between English and over 24 European languages.[22] This unlocks a powerful capability for the factory: automated localization. A video created in English can be automatically captioned in Spanish or German without additional models.

VRAM Impact:
At 1 billion parameters, Canary-1B is lightweight. In FP16, it consumes roughly 2-2.5 GB of VRAM.25 This modest footprint allows it to be loaded effortlessly after the heavy VLM is unloaded.

5.2 The Importance of Forced Alignment

Standard ASR models output text with rough timestamps. For high-energy shorts, captions need to pop exactly when the word is spoken. This requires **Forced Alignment**. The industry standard has shifted away from the complex, Kaldi-based Montreal Forced Aligner (MFA) toward **CTC-Segmentation** approaches.[26]

Tools like **Wav2TextGrid** or the **TorchAudio forced alignment API** leverage the CTC (Connectionist Temporal Classification) probabilities from models like Wav2Vec2 or Canary itself to align the transcript to the waveform.[27] This process is extremely fast and VRAM-efficient. By using a specialized alignment model like mms-300m-1130-forced-aligner (which supports 158 languages), the factory can achieve phoneme-level precision with minimal resource overhead.[29] The alignment step typically runs on the CPU or consumes negligible GPU memory (<500MB), ensuring it never bottlenecks the pipeline.

6. The Artist Agent: Generative Video and B-Roll

The final and most computationally demanding stage is the generation of "b-roll"—supplemental footage to cover cuts, visualize spoken concepts, or add visual variety. This requires a Text-to-Video (T2V) diffusion model. In the 16GB VRAM era, this was previously impossible without massive quality compromises. The 2025-2026 wave of models has changed this.

6.1 Wan 2.1: The Low-VRAM Miracle

The standout model for this task is **Wan 2.1**, specifically the **1.3B** parameter variant, released by Alibaba.[30] While a 14B version exists for enterprise clusters, the 1.3B model was explicitly architected for consumer GPUs. It requires only **~8.19 GB of VRAM** to run in full FP16 precision.[32]

Despite its small size, Wan 2.1 1.3B delivers shocking quality. It utilizes a novel **3D Causal VAE** (Variational Autoencoder) that compresses video data efficiently while preserving temporal

continuity.[33] It supports Text-to-Video generation at 480p or 720p resolutions—perfectly adequate for the small screens where shorts are consumed.

Comparative Advantage:
Comparing Wan 2.1 to its peers highlights its suitability for the RTX 5080:

- **HunyuanVideo:** While arguably the quality king (rivaling Runway Gen-3), it is a massive model. Even heavily quantized GGUF versions (Q4) push the 16GB limit to the breaking point, often leaving no room for the OS overhead and causing instability.[34]
- **CogVideoX-2B:** This model is another viable option, with FP16 inference requiring ~10-12GB (using diffusers optimizations like tiling).[36] However, Wan 2.1 1.3B offers a better quality-to-VRAM ratio and includes robust support for Chinese and English prompts.[32]
- **LTX-Video:** A 2B parameter model that emphasizes speed ("faster than real-time"). It is an excellent secondary choice if throughput is the priority over visual fidelity.[37]

For the autonomous factory, **Wan 2.1 1.3B** is the recommended workhorse. Its 8GB footprint fits comfortably within the 16GB budget, leaving ample headroom (roughly 6-8GB) to prevent OOM crashes, even if the system is running a display or background tasks.

6.2 Quantization in Generation

For users daring enough to try the larger **Wan 2.1 14B** or **HunyuanVideo**, **GGUF quantization** is the only path forward. Libraries like **ComfyUI** with GGUF nodes allow these massive models to be loaded in 4-bit precision.[38] However, generating video with a quantized 14B model on a single 5080 is slow—often taking minutes per clip—which may bottleneck the "factory" throughput. The 1.3B model maintains the "assembly line" speed.

7. System Architecture and VRAM Orchestration

The successful deployment of these models relies on a rigid orchestration strategy. The pipeline must be viewed as a state machine where only the necessary assets are resident in VRAM.

7.1 The Sequential VRAM Budget

The following breakdown illustrates the memory state at each phase of production on the RTX 5080:

Table 1: Operational VRAM States

Phase	Active Model	Model Type	Precision/Format	Est. VRAM Load	System Overhead	Total Usage	Margin (16GB)

1. Analysis	Qwen2.5-VL-7B	VLM	GGUF Q4_K_M	~6.5 GB	~2.0 GB	~8.5 GB	**Safe**
2. Audio	Canary-1B-v2	ASR	FP16	~2.2 GB	~2.0 GB	~4.2 GB	**Very Safe**
3. Tracking	SAM 2 (Small)	Tracker	FP16	~1.8 GB	~2.0 GB	~3.8 GB	**Very Safe**
4. Generation	Wan 2.1 (1.3B)	DiT	FP16	~8.2 GB	~2.0 GB	~10.2 GB	**Safe**

Note on Phases 2 & 3: The VRAM usage in the Audio and Tracking phases is low enough that they could potentially run continuously or in parallel. However, the transition to Phase 4 (Generation) requires a "hard flush" of VRAM. The orchestration script must explicitly unload the ASR and Tracking models, trigger a CUDA cache clear (torch.cuda.empty_cache()), and then load the Video Generation model.

7.2 The Agentic Workflow

The "glue" holding these models together is the JSON output from Qwen2.5-VL. The VLM acts as the brain.

1. **Ingest:** Qwen views the video.
2. **Plan:** Qwen outputs:
   JSON
   ```
   {
     "segments": [
       {
         "start": "00:45",
         "end": "01:15",
         "virality_reason": "Humorous reaction",
         "subject_to_track": "man in blue hoodie",
         "b_roll_prompts": ["cinematic shot of a coding setup", "cyberpunk city rain"]
       }
     ]
   }
   ```
3. **Execute:** The Python orchestrator parses this JSON.

- It calls **Canary** to transcribe 00:45-01:15.
- It initializes **SAM 2** with the prompt "man in blue hoodie" to generate crop coordinates.
- It uses **FFmpeg** to cut and crop the video based on SAM 2's data.
- If the visual quality is low, it loads **Wan 2.1** to generate the "cyberpunk city rain" clip and overlays it.

8. Conclusion

By late 2025, the open-source community has successfully engineered a suite of models that dismantle the hardware barriers to entry for AI video production. The NVIDIA RTX 5080, with its 16GB of VRAM, is no longer just a gaming card; it is a viable industrial platform for an autonomous content factory.

The recommended stack is clear:

- **Video Understanding: Qwen2.5-VL-7B-Instruct (GGUF)** offers the best balance of agentic reasoning and temporal precision.
- **Visual Tracking: SAM 2** provides cinema-grade subject isolation, optimized via streaming memory to run on consumer cards.
- **ASR: NVIDIA Canary-1B-v2** delivers SOTA multilingual transcription in a lightweight package.
- **Generation: Wan 2.1 (1.3B)** is the breakthrough enabling high-quality video synthesis within an 8GB envelope.

This architecture shifts the value proposition from hardware ownership to pipeline orchestration. The capability to produce viral shorts autonomously is now defined not by who owns the biggest cluster, but by who can best integrate these quantized, specialized agents into a cohesive, intelligent workflow.

Works cited

1. Qwen3-VL is the multimodal large language model series developed by Qwen team, Alibaba Cloud. - GitHub, accessed January 17, 2026, https://github.com/QwenLM/Qwen3-VL
2. Qwen/Qwen2.5-VL-7B-Instruct - Hugging Face, accessed January 17, 2026, https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct
3. Video-MME: The First-Ever Comprehensive Evaluation Benchmark of Multi-modal LLMs in Video Analysis - GitHub, accessed January 17, 2026, https://github.com/MME-Benchmarks/Video-MME
4. Need suggestions on how to load the qwen2-vl-7b-instruct model without quant smoothly in the local setup. - Reddit, accessed January 17, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1h342kb/need_suggestions_on_how_to_load_the/
5. A formula that predicts GGUF VRAM usage from GPU layers and context length,

accessed January 17, 2026, https://oobabooga.github.io/blog/posts/gguf-vram-formula/
6. Expanding Performance Boundaries of Open-Source Multimodal Models with Model, Data, and Test-Time Scaling - arXiv, accessed January 17, 2026, https://arxiv.org/html/2412.05271v2
7. InternVL2.5: Expanding Performance Boundaries of Open-Source Multimodal Models with Model, Data, and Test-Time Scaling, accessed January 17, 2026, https://internvl.github.io/blog/2024-12-05-InternVL-2.5/
8. Quick Start of InternVL 2.5 Series, accessed January 17, 2026, https://internvl.readthedocs.io/en/latest/internvl2.5/quick_start.html
9. minicpm-v - Ollama, accessed January 17, 2026, https://ollama.com/library/minicpm-v
10. MiniCPM-V 4.5: A GPT-4o Level MLLM for Single Image, Multi Image and High-FPS Video Understanding on Your Phone - GitHub, accessed January 17, 2026, https://github.com/OpenBMB/MiniCPM-V
11. MiniCPM-o 2.6: An 8B size, GPT-4o level Omni Model runs on device - Reddit, accessed January 17, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1i11961/minicpmo_26_an_8b_size_gpt4o_level_omni_model/
12. SAM 2: Segment Anything Model 2 - Ultralytics YOLO Docs, accessed January 17, 2026, https://docs.ultralytics.com/models/sam-2/
13. Improvement of SAM2 Algorithm Based on Kalman Filtering for Long-Term Video Object Segmentation - NIH, accessed January 17, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12252479/
14. facebookresearch/sam2: The repository provides code for running inference with the Meta Segment Anything Model 2 (SAM 2), links for downloading the trained model checkpoints, and example notebooks that show how to use the model. - GitHub, accessed January 17, 2026, https://github.com/facebookresearch/sam2
15. Det-SAM2: Self-Prompting Segmentation Framework Based on Segment Anything Model 2, accessed January 17, 2026, https://arxiv.org/html/2411.18977v1
16. VRAM requirements/benchmarks? · Issue #118 · facebookresearch/sam2 - GitHub, accessed January 17, 2026, https://github.com/facebookresearch/sam2/issues/118
17. YOLO-World: Real-Time, Zero-Shot Object Detection - Roboflow Blog, accessed January 17, 2026, https://blog.roboflow.com/what-is-yolo-world/
18. AILab-CVC/YOLO-World: [CVPR 2024] Real-Time Open-Vocabulary Object Detection - GitHub, accessed January 17, 2026, https://github.com/AILab-CVC/YOLO-World
19. Paper page - CAMELTrack: Context-Aware Multi-cue ExpLoitation for Online Multi-Object Tracking - Hugging Face, accessed January 17, 2026, https://huggingface.co/papers/2505.01257
20. A Comprehensive Guide to Object Tracking Algorithms in 2025 | Datature Blog, accessed January 17, 2026, https://datature.com/blog/a-comprehensive-guide-to-object-tracking-algorithms-in-2025

21. Riva | Speech and Translation AI - NVIDIA, accessed January 17, 2026, https://www.nvidia.com/en-us/ai-data-science/products/riva/
22. nvidia/canary-1b-v2 - Hugging Face, accessed January 17, 2026, https://huggingface.co/nvidia/canary-1b-v2
23. New Standard for Speech Recognition and Translation from the NVIDIA NeMo Canary Model, accessed January 17, 2026, https://developer.nvidia.com/blog/new-standard-for-speech-recognition-and-translation-from-the-nvidia-nemo-canary-model/
24. Speech Recognition/Translation: Canary - NGC Catalog - NVIDIA, accessed January 17, 2026, https://catalog.ngc.nvidia.com/orgs/nvidia/teams/riva/models/canary-riva-1b
25. Best open source speech-to-text (STT) model in 2026 (with benchmarks) | Blog - Northflank, accessed January 17, 2026, https://northflank.com/blog/best-open-source-speech-to-text-stt-model-in-2025-benchmarks
26. Performance of Forced-Alignment Algorithms on Children's Speech - PMC - NIH, accessed January 17, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8740721/
27. Forced Alignment with Wav2Vec2 — Torchaudio 2.9.0 documentation, accessed January 17, 2026, https://docs.pytorch.org/audio/stable/tutorials/forced_alignment_tutorial.html
28. pkadambi/Wav2TextGrid: Speaker adaptive forced alignment (phonetic segmentation) using Wav2Vec2 - GitHub, accessed January 17, 2026, https://github.com/pkadambi/Wav2TextGrid
29. MahmoudAshraf/mms-300m-1130-forced-aligner - Hugging Face, accessed January 17, 2026, https://huggingface.co/MahmoudAshraf/mms-300m-1130-forced-aligner
30. Qwen's Wan 2.1: A Guide With Practical Examples - DataCamp, accessed January 17, 2026, https://www.datacamp.com/tutorial/wan-2-1
31. Wan-AI/Wan2.1-T2V-1.3B - Hugging Face, accessed January 17, 2026, https://huggingface.co/Wan-AI/Wan2.1-T2V-1.3B
32. Wan-Video/Wan2.1: Wan: Open and Advanced Large-Scale Video Generative Models - GitHub, accessed January 17, 2026, https://github.com/Wan-Video/Wan2.1
33. replicate/cog-wan-2.1: Wan: Open and Advanced Large-Scale Video Generative Models - GitHub, accessed January 17, 2026, https://github.com/replicate/cog-wan-2.1
34. tencent/HunyuanVideo-1.5 - Hugging Face, accessed January 17, 2026, https://huggingface.co/tencent/HunyuanVideo-1.5
35. HunyuanVideoGP V5 breaks the laws of VRAM: generate a 10.5s duration video at 1280x720 (+ loras) with 24 GB of VRAM or a 14s duration video at 848x480 (+ loras) video with 16 GB of VRAM, no quantization : r/StableDiffusion - Reddit, accessed January 17, 2026, https://www.reddit.com/r/StableDiffusion/comments/1iybxwt/hunyuanvideogp_v5_breaks_the_laws_of_vram/
36. Text-to-video generation: CogVideoX (2024) and CogVideo (ICLR 2023) - GitHub,

accessed January 17, 2026, https://github.com/siliconflow/CogVideo-P
37. Choosing the Best Open-Source Video Generation Model - WhiteFiber, accessed January 17, 2026, https://www.whitefiber.com/blog/best-open-source-video-generation-model
38. Run LTX Video on Low VRAM! Fast GGUF Workflow & Audio Noise Fix for ComfyUI, accessed January 17, 2026, https://www.youtube.com/watch?v=r7TACD329wY